Advanced Bio-Electronics

Pulmonary Functions

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Abstract

Pulmonary function tests are valuable investigations in the management of patients with suspected or previously diagnosed respiratory disease. They aid diagnosis, help monitor response to treatment and can guide decisions regarding further treatment and intervention [1]. In this review, will be presented three different pulmonary function tests, as each one examines an unique angle of the respiratory system. The purpose of the first test is to compare between normal values of pulmonary volumes and capacities of subjects differing in sex, age and height. For testing findings' reliability, it was compared to the theoretical values. The spirometry results showed unsuitability of VC and RV values to the theoretical ones, as a neglectable ERV volume detected and the RV weren't calculated in the experiment. The purpose of the second part of the experiment is to examine the behavior of respiratory signals during different simulations. The simulations include normal breathing, hyperventilation, hypoventilation and respiratory during coughing and reading. The signal in this part was recorded and measured using pneumography and air temperature transducers.The respiratory results were mostly adequate with the expectation and clearly showed difference between the different breathing functionalities that were tested. The purpose of the third part of the experiment is to examine an application for mobile phones which can measure FEV1 and FVC values and the ratio between them to diagnose COPD. The results show that the mobile phones manage to well diagnose unpresence of disease. For future experiments, the improvement recommendation is to apply the application on a diverse group of examinees and.

1. INTRODUCTION

The respiratory system is a biological system consisting of the nose, pharynx, larynx, trachea, bronchi, bronchioles, and lungs [2]. The pulmonary system's ventilation is activated by three pressure types - the atmospheric pressure (P_{atm}), the alveolar pressure (P_{alv}) and the intrapleural pressure (P_{ip}) [3]. The respiratory cycle consists of inhalation and exhalation, when in inhalation, the skeletal muscles contract, and thus the volume of the thoracic cavity increases [4]. The increase in volume causes the alveolar pressure to get under the atmospheric pressure. Therefore, a pressure gradient is produced which causes the air (with the oxygen) to come into the lungs [3]. After that, the oxygen moves into the blood through the capillaries, and the hemoglobin, which is a protein in the red blood cells, carries the oxygen around the body. The cells use the oxygen during the metabolic process, convert it into energy and produce waste called carbon dioxide [5]. Carbon dioxide can form carbonic acid and dissociates to H+ and bicarbonate, and that can decrease the blood's pH [6]. Thus, there is a paramount importance to clean this waste out of the body (by the blood circulation) and to control the respiratory cycle [5]. The respiratory cycle is controlled by chemoreceptors that monitor the pH in the blood and the partial pressure of carbon dioxide in the blood [7], and by baroreceptors - receptors which monitor the degree of stretch in the lungs and indicate the lung volume [8].

Respiratory rates change based on many health and activity factors such as: resting, exercising, panicking, eating, and even illness [9]. During exercise, two of the important organs of the body come into action: the heart and the lungs. As mentioned, the lungs bring oxygen into the body, to provide energy, and remove carbon dioxide. The heart pumps the oxygen to the muscles that are doing the exercise. The muscles work harder, and the body uses more oxygen and produces more carbon dioxide. To cope with this extra demand, the breathing has to increase from about 15 times a minute (12 liters of air) when you are resting, up to about 40–60 times a minute (100 liters of air) during exercise. The circulation also speeds up to take the oxygen to the muscles so that they can keep moving [10].

Hypoventilation and hyperventilation are two abnormal respiratory conditions in which breathing is not performed properly. Hypoventilation is the abnormal retention of carbon dioxide in the blood due to the poor exchange of carbon dioxide and oxygen within the lungs. Slow and shallow breathing characterizes this breathing disorder. With Hypoventilation, the body cannot remove carbon dioxide efficiently, and as a result, there is a build-up of carbon dioxide with little oxygen available for the functioning of the body. Hypoventilation differs from hyperventilation. Hyperventilation is when someone breathes too fast, and exhales more than he takes in [11]. Chemoreceptors, in turn, respond to pH changes as they become more acidic and send sensory input to the brain to stimulate hyperventilation. The result is a slow and deep breathing pattern that helps eliminate carbon dioxide from the body. Likewise, when arterial PCO2 drops, pH in the cerebrospinal fluid becomes alkalotic, and hypoventilation ensues. Therefore, arterial PCO2 is the chief determinant of the respiratory drive under normal conditions [12].

Pulmonary function tests enable accurate and reproducible assessment of the functional state of the respiration system, could indicate different patterns of abnormalities, provide a better baseline estimation of prognosis, and be used to monitor disease progression and therapy response. But those tests

cannot diagnose specific diseases [13]. The main ventilatory patterns measured by pulmonary function tests are obstruction and restriction. Categorization of these patterns is dictated by the lung volumes, both static and forced [13].

Spirometry is a basic lung function test that measures the air that is expired and inspired. There are three basic related measurements: volume, time and airflow. Spirometry is objective, noninvasive, sensitive to early change and reproducible. With the availability of portable meters, it can be performed almost anywhere, and with the right training, it can be performed by anybody. It is performed to detect the presence or absence of lung disease, quantify lung impairment, monitor the effects of occupational/environmental exposures, and determine the effects of medications [14].

Spirometry measures include the following volumes and capacities:

Table 1: Lung Volumes and Capacities

Tidal Volume (TV)	The volume of air either inspired of expired during tidal respiration [15].
Inspiratory Reserve Volume (IRV)	The volume of air inspired by forcible inspiration after tidal respiration [15].
Expiratory Reserve Volume (ERV)	The volume of air expired by forcible inspiration after tidal respiration [15].
Residual Volume (RV)	The volume of gas remaining within the lungs at the end of maximum expiration [15].
Total Lung Capacity (TLC)	Gas contained within the lungs after a maximal inspiration [15] .
Inspiratory Capacity (IC)	Gas contained within the lungs after maximal [15].
Functional Residual Capacity (FRC)	The volume of gas remaining in the lungs at the end of tidal exhalation [15].
Expiratory capacity (EC)	The maximum volume of air that can be exhaled after a normal resting inhalation [16].
Vital capacity (VC)	The maximum volume of air that can be exhaled starting from maximum inspiration [17].
Functional residual capacity (FRC)	The volume of air in the lungs following a tidal volume exhalation [17].

During the test, A graph (Figure 1) is obtained that describes the volumes within the lungs while a person's breathing. The graph can be analyzed, and different volumes obtained from it can be compared to known values of volumes in the normal range to compare the functionality of the patient's lung activity in relation to a healthy person with the same characteristics [16].

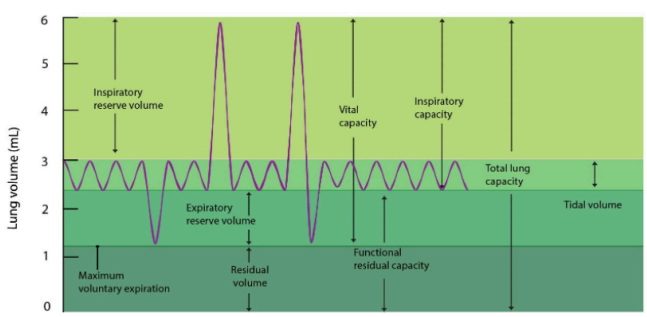

Figure 1: static lung volumes and capacities which can be determined in a spirometry test *[16]*

Another approach of spirometry test can measure two more values: FEV1- Forced expiratory volume in one second; FVC - Forced vital capacity exhaled with effort in a complete breath [18]. COPD is typically diagnosed using spirometry, based on FEV1/FVC ratio. Chronic Obstructive Pulmonary Disease (COPD) is a common progressive lung disease that causes difficulty in breathing. If the FEV1/FVC ratio is found to be abnormal, the next step is grading the abnormality to determine the severity of the condition. The American Thoracic Society has set specific guidelines for this purpose (table 2) [19]:

Table 2: Abnormal FEV1 and FVC Results [19]

FEV1/FVC ratio [%]	Severity of Condition
>70	Mild
60-69	Moderate
50-59	Moderately Serve
35-49	Serve
<34	Vert Serve

A normal ratio is considered to be 70% to 80% in adults, and 85% in children [19].

A new approach for diagnosing COPD using mobile phones is proposed. The approach takes advantage of the built-in microphone and proximity sensor in correctly recording the user's exhalation [18].

Lung function, such as vital capacity, reference values are traditionally based on anthropometric factors, such as weight, height, sex, and age. Several parameters (TLC, VC, RV, FVC and FEV1) are affected by height since they are proportional to body size. This means that in tall individuals, who accordingly have greater lung capacity, lung volume will decrease at a greater rate to shorter individuals as they grow older [20]. In addition, very high or very low weights are harmful to Lung function [20]. Moreover, standard morphometric methods confirm that among individuals of the same weight and height, men have larger lungs than women, and consequently, a larger number of bronchi, greater alveolar surface area, and a wider airway caliber [20]. Furthermore, age has historically been one of the major factors in the evaluation of lung function. As much as getting older, the lung function is lower [20].

The experiment is divided into three parts: calculating and comparing pulmonary volumes and capacities; measuring and recording ventilation using pneumography, air temperature transducers and chest expansion; testing an application for mobile phones which can measure FEV1/FVC ratio for diagnosing COPD.

2. METHODS

The experiment included three types of pulmonary function tests: two versions of spirometry test and test which was measured using pneumography and air temperature transducers. The first and third parts were examined at a 26 years old 161 cm height female, and the second part was examined at a 25 years old 168 cm height female. In all the parts of the experiments the examinee was seated at rest. There is a significant importance of maintaining rest while performing pulmonary tests due to the fact that the breathing activity is much influenced by the physical situation in which an individual is at. Therefore, in order to obtain as much as possible accuracy and uniformity in the results, it is very important to enable resting situations during the experiment and keep on a constant stage as possible.

2.1. Part 1

In the first part, a spirometer was used. This device calculates the differences in the air volume while breathing into it. This part aimed to extract the lung volumes and capacities which can be determined in a spirometry test, and also to compare the observed values of volume and capacity with average values and normal pulmonary volumes. Airflow represents the

dynamic movement of the air while breathing, and thus at the moment that exhaling starts the airflow graph immediately decreases while the air volume is still above the average breathing volume. By this way, airflow can describe the changes in lung volume.

2.1.1 Equipment & Materials

For the first part of the experiment:

- BIOPAC Airflow Transducer (SS11LA)
- BIOPAC Bacteriological Filter (AFT1)
- BIOPAC Disposable Mouthpiece (AFT2)
- BIOPAC Nose clip (AFT3)
- BIOPAC student lab software V3.0
- BIOPAC Calibration Syringe (AFT6)
- BIOPAC acquisition unit (MP30)
- BIOPAC wall transformer (AC100A)
- BIOPAC serial cable (CBLSERA)

2.1.2. Experimental Procedure

In the first step of the experiment, a calibration procedure was performed. The calibration included two stages; measuring the changes without connecting the device to anything as the syringe was all the way out and was held upright (because of the transducer's sensitivity to gravity) for 8 seconds. Then, measuring the changes while pulling and pushing the syringe plunger in and out for 10 strokes to ensure the result graphs behaved in the expected way. The calibration, as also the experiment itself, was performed using a filter which connected to the syringe and the whole syringe-filter assembly was inserted into the airflow transducer. After calibration, the experiment was started; the examinee was a 26-year-old young woman with 161cm height. The examinee sat on a chair with a nose clip closing her nostrils so she could breathe only from her mouth. The test first started with a set of five regular inhales and exhales. Right afterwards, a single extreme inhale and a single extreme exhale were performed and then again, a set of five regular breathing. All the results were saved as a MATLAB file.

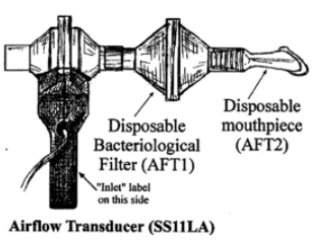

Figure 2: spirometer machine

2.1.3. Data Analysis and Values Calculations

The sampling time given by the Biopack was dT = 0.01 sec, so the sampling frequency (Fs) was calculated according to the equation:

$$(1) \quad Fs = \frac{1}{dT}\ [Hz]$$

To calculate the spirometry parameters MATLAB's built-in functions and the following formulas [21] were used.

The RV value was a default value that was given by the Biopack system and was 1 Liter. The TLC value is calculated as the maximum value of the overall signal's volume. The TV value calculated using the "findpeaks" MATLAB's function - after detecting the signal's peaks and minimum values, the maximum and minimum values of the intense inspiration part was removed and then the TV calculated as the difference between the maximum and the minimum volume of the moderate breathing parts in the overall record. The ERV calculated from the difference between the minimum TV value and RV value. The other values calculator based on these results using the equations below:

(2) VC [L]= (TLC - RV) [L]

(3) EC [L]= (ERV + TV) [L]

(4) IC [L]= (VC - ERV) [L]

(5) IRV [L]= (IC - TV) [L]

(6) FRC [L]= (TLC - IC)[L]

When the values explain in table (1).

The theoretical values calculated based on the preparation to the experiment [21] where vital capacity in liters for females (since the examinee was a female). is calculated by the equation:

(7) VC_theoretical = 0.041*H - 0.018*A - 2.69 [L]

Residual volume in liters for females is calculated by the equation:

(8) RV_theoretical = (21.6*H + 20.7*A - 2840)/1000 [L]

When A = 26 (the examinee's age), and H = 161 (the examinee's height).

In order to compare the theoretical values to the values that was measured in the experiment a relative error value was calculated according to the equation:

$$(9) \quad Error = 100 \cdot \left| \frac{x_{theory} - x_{measured}}{x_{theory}} \right| [\%]$$

2.2. Part 2

The second part of the experiment was performed using a breast belt with a pneumography transducer which measures the changes in force applied on the device as the breast expands and shrinks while breathing. As described in the introduction section, the changes in breast activity has a major role in the breathing cycle as when the breast expands there is a difference in air pressure between the lung and the outside pressure and thus air flows inside the lungs, and vice versa. (meaning changing lung values). Simultaneously, a thermistor transducer was used next to the examinee's nostril to measure the changes in temperature while breathing. The inhalation is accompanied by insertion of cold air while exhalation is accompanied by ejection of hot air, when voltage shows increasing, which is caused by inhale, the temperature expected to show decrease, which is caused by cold air that is detected in the temperature sensor that located under the examinee's nostril. Thus, temperature results can express the lungs' activity and to evaluate their changes in volume (meaning changing lung values).

This experimental part measured a few different breathing stages: normal breathing (eupnea), hyperventilation, hypoventilation, coughing and reading. This part aimed to measure ventilation utilizing pneumography and air temperature transducers, show the connection between ventilation and temperature changes in airflow and understand the influence of cerebral and chemoreceptors on the rate and depth of the breathing medullary control centers.

Even Though, every segment in the experiment is taken at a different time, it is still possible to make a comparison between the segments, and that can be explained by several reasons. First, the experiment was executed at the same person within a short time distance between one segment to another, and thus, it is much likely to assume that the examinee's conditions did not change so much between the segments and so, each segment has the same base of assumptions and it minimizes the optional errors. Despite that, there was a break between the segment when the examinee rested and regulated the breathing, and due that it can be assumed that each segment was not very much affected by the previous segment's physiological activity. In addition, the experiment aimed to show a general trend and behavior of the breathing in different situations which has a major influence over the results, and thus the fact that each segment was taken at a different time shouldn't have a dramatic change on the general results trend nor the behavior.

2.2.1 Equipment & Materials

For the second part of the experiment:

- BIOPAC Respiratory Transducer SS5LB
- BIOPAC Temperature Transducer SS6L
- BIOPAC single sided (surgical) tape (TAPE1)
- Chair without armrests
- BIOPAC Student Lab System: BSL4 software, MP36
- Computer System

2.2.2. Experimental Procedure

First, the Respiratory and the temperature devices were plugged into channels 1 and 2 respectively. Then, the respiratory transducer adjusted around the examinee's chest, below the armpits and above the nipples. Next, the temperature transducer was held under the examinee's nostril (figure 3) and then the calibration procedure began. First, recording while the examinee seated and breathing normally though his nose for 8 seconds. Both channels should show graphs which behave similar to the expectation. The experiment procedure was divided into 5 different segments: First, eupnea segment when the examinee performed normal resting breathing for 20 seconds. Then, hyperventilation breathing was mimicked for 33 seconds. Afterwards, hypoventilation was mimicked for 28 seconds. Then, coughing during a normal resting breathing for a segment duration of 42 seconds. At the end a breathing while reading text segment was recorded for about 29 seconds.

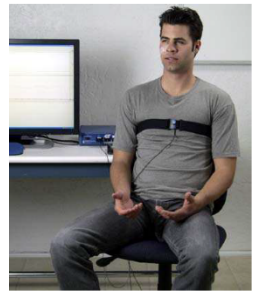

Figure 3: position of the examinee in the second part

2.2.3. Data Analysis and Values Calculations

The sampling frequency (Fs) was calculated the same as the first part (formula (1)).

The mean value was calculated using the equation:

$$(10)\ mean = \frac{1}{N}\sum_{i=1}^{N} x_i$$

When N is the number of objects in the sample; x_i is the value of object i in the sample.

The standard deviation value was calculated using the equation:

$$(11)\ \sigma = \sqrt{\frac{\sum_{i=1}^{N}\left(x_i - \underline{x}\right)^2}{N-1}}$$

When N is the number of objects in the sample; x_i is the value of object i in the sample; $\underline{x}$ is the mean value of the sample.

The Respiration Rate (RR) calculated according to the equation:

$$(12) \qquad RR = \frac{60}{\Delta t}\left[\frac{BREATHS}{MINUTE}\right]$$

When Δt is the time difference between two adjacent peaks of the respiration signal.

The RR value presents the respiration rate that would have been achieved if the rate would have been based on the rate that was calculated between two adjacent peaks in the graph. In order to calculate this parameter, it was needed to calculate the duration time between two adjacent peaks and then calculate how many peaks would have been detected during 1 minute if the rate was constant. The time difference between two adjacent peaks in the signal was calculated using the built-in "findpeaks" MATLAB function with the 'MinPeakDistance' feature in changing values that were set for each segment in that order; eupnea: 300 minimum indices distance between the peaks, hyperventilation: 100 minimum indices distance between the peaks, hypoventilation: 500 minimum indices distance between the peaks, coughing: 300 minimum indices distance between the peaks, reading: 400 minimum indices distance between the peaks (every 1 second was represented by 100 indices). Then each RR value was kept in a vector in order to display the RR graph as a function of time next to the respiration voltage graph.

To find the correspondence between the hypoventilation and hyperventilation to the coughing-reading segment the built-in

"wcoherence" function was used with ,'FrequencyLimits' of 0 to 5 Hz, as frequencies above are probably refers only to noises in the signal.

2.3. Part 3

The third part of the experiment includes a control group and a test group. The control group is based on another version of spirometry test, which measures the values FEV1 and FVC. The test group is a new approach which uses the built-in microphone of mobile phones and a proximity sensor to measure those values [18]. A comparison was performed between the resulting values as well as theoretical values. The main goal of this part is to calculate the FEV1/FVC ratio, which enables diagnosing COPD (chronic inflammatory lung disease). Because FEV1 and FVC values depend on anthropometric factors, the three methods (spirometry test, mobile phone, theoretical calculation) were considered.

2.3.1 Equipment & Materials

Control group - The equipment for the spirometry test is identical to the equipment of the first part of the experiment.

Test group - The equipment for the test group in the third part includes an Android mobile phone with a dedicated application. The application is free as it doesn't require anything but the mobile phone and its built-in sensors (such as external servers) [18].

2.3.2. Experimental Procedure

Control group - The first step of a calibration procedure was skipped based on the calibration of the first part. The examinee was the examinee of the first part. The examinee sat on a chair with a nose clip closing her nostrils so she could breathe only from her mouth. This part was started with a signal three sets of regular inhales and exhales, which did not record. Right afterwards, a maximum inspiration followed by forced expiration to squeeze the air out of the lungs, when only the expiration was recorded. The results were saved as a MATLAB file.

Test group - This part does not include a calibration procedure. The rest of the procedure is identical to the control group while the distance between the examinee's face to the mobile phone is at least 5 cm. This distance is measured by the proximity sensor in the phone - an interrupt-based sensor, which gives only two values: 1 or 0. It responds to a change that is more than 5 cm away from the examinee's face - If the mobile phone is placed at a distance less than or equal to 5 cm from the examinee's face, no readings will be taken. As the phone moves away from the user's face for more than 5 cm, the sensor gets triggered to indicate to the user to hold the phone at that distance from the face and start the exhaling process [18].

2.1.3. Data Analysis and Values Calculations

Control group - The information that received from the control group processed using MATLAB, when:

- FEV1 is the volume which is received after one second from the start of the forced expiration (meaning from the start of the recording).

- FVC is the last volume of the measure. This value should belong to the plateau part of the graph which presents the result of this test and should be the maximum volume of the measure.

After extracting these two values, the ratio between them is calculated by the following formula:

$$(13) \qquad FEV_1\% = \frac{FEV}{FVC} \cdot 100\ [\%]$$

According to the value, an assessment is made on diagnosing COPD.

Test group - After the exhalation is recorded locally on the phone, it is analyzed, and the COPD diagnosis is assessed. Meaning, the user gets four values at the end of the test: FEV1, FVC, FEV1/FVC ratio and COPD diagnosis. The data is also securely sent to a remote server using Wi-Fi or 3G connection and stored in the server's database. That is for providing medical feedback and advice (figure 4) [18].

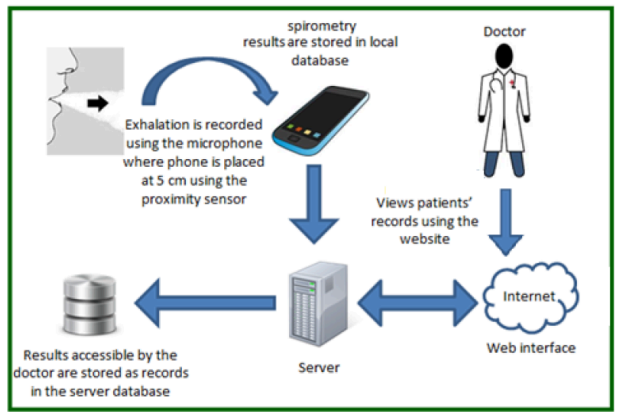

Figure 4: COPD Diagnosis System Overview [18]

In order to diagnose COPD from the user's exhalations, the loudest frequencies need to be computed, because the air blown to the microphone produces the loudest frequency. The Android mobile application is developed in JAVA using Eclipse IDE with the Android Development Tools plugin. The algorithm performs as follows: the exhalations are recorded at a sampling rate of 44.1 kHz using the built-in microphone of the phone and then analyzed by dividing the signal into windows of size 0.1 seconds with a 50% overlap. Discrete Fourier Transform (DFT) is performed on each window to find the power density spectrum. The peak magnitude of each window is plotted which represents the flow of air exhaled in L/s [18].

Theoretical Values

The theoretical values calculate based on the following formulas:

$$(14) \quad FEV_1 = Race \cdot 1.08 \cdot [(0.0395 \cdot Hight) - (0.025 \cdot Age) - 2.6]$$

$$(15) \quad FVC = Race \cdot 1.15 \cdot [(0.0443 \cdot Hight) - (0.026 \cdot Age) - 2.89]$$

When Race = 1 [22].

Comparison (Control, Test & Theoretical values)

The errors between all the relevant values and results are calculated due to formula (9).

3. RESULTS

3.1 Part 1

3.1.1 Spirometry Test

Table 3: volumes & capacities from the first part in Liters

TV	IRV	ERV	RV	IC	EC	FRC	VC	TLC
0.281	1.494	0.041	1	1.775	0.322	1.041	1.817	2.817

Table 3 presents the resulting values from the experiment. As it can be seen in the graph, the RV and FRC have very close values with approximately 4% relative error, and correspondingly ERV is very low. Also, VC and IC have very close values with approximately 4% relative error.

Table 4: theoretical values & experiment results

	RV [L]	VC [L]
Theoretical Value	1.176	3.443
Experiment Value	1	1.817
Error [%]	14.952	47.236

The theoretical values of RV and VC were calculated by formulas (7,8). The relative errors calculated (formula 9) are larger than 5% (table 4). Both vital capacity and residual volume which were measured, are smaller than the theoretical values.

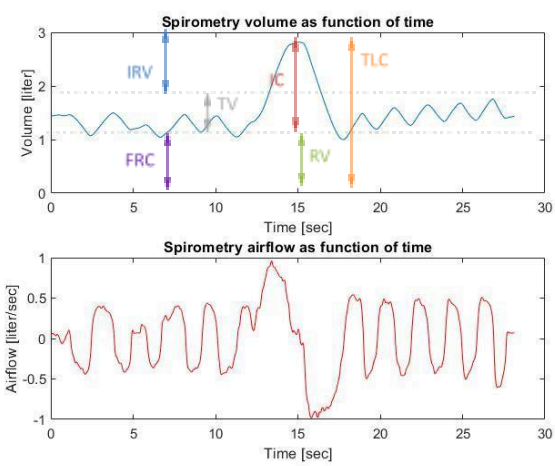

Figure 5: Spirometry test signals of volume [Liter] and airflow [Liter/sec] as function of time

Note: Some of the volumes were not mentioned in the spirometry volume graph as their appearance can cause confusion with the other volumes by the fact that the extreme exhale cannot be clearly seen in the graph.

In the graph in figure 5 above, a moderate breath can be seen in the first and the last 10 seconds as expected in resting breathing. While in the 15 second from the beginning of the test it's clear that an intensive inhale occurred. It can clearly be seen that in the beginning of the extreme inhalation the air flow increased and afterwards decreased. It must be mentioned that no intense exhalation can be seen in the graph, though the second resting phase seems to increase volume with proportional inhale and exhale ratio.

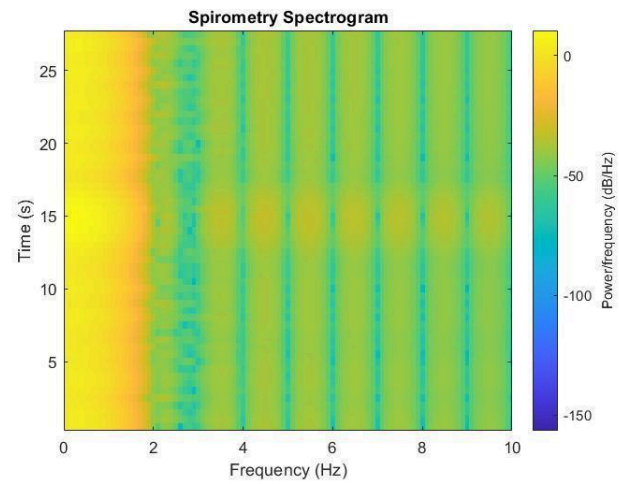

Figure 6: Spectrogram of spirometry volume signal in time [sec] as function of frequency [Hz]

The spirometry spectrogram in figure 6 shows a high power per frequency for the frequency beneath 2 Hz at all times. In the first and last 10 seconds of the spirometry test there is a low frequency breath of 5 Hz. Also, a high power can be seen around the 15th second from the beginning of the test, in all the frequencies.

Part 2

3.1.2 Pneumography Transducer & Thermistors

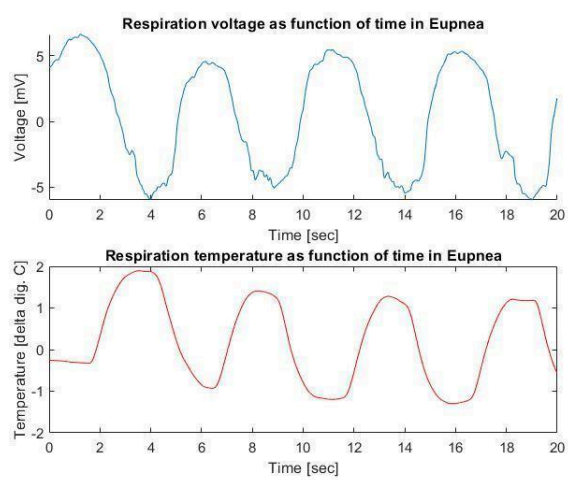

Figure 7: Voltage [mV] and Temperature [ΔC°] signals vs *time of Respiration test in eupnea*

As shown in figure 7 above, the temperature's signal has an inverse trend to the respiratory voltage trend. In addition, the respiratory rate looks constant and moderate.

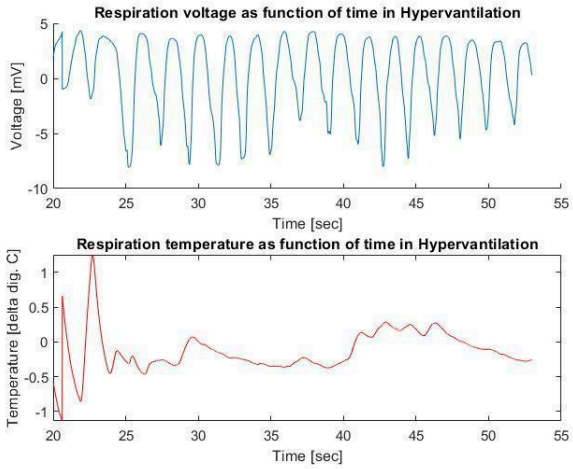

Figure 8: Voltage [mV] and Temperature [ΔC°] signals vs *time of Respiration test in hyperventilation*

Figure 8 above shows a high respiratory rate with an uncertain temperature trend. The deepness of the respiration that is shown in the voltage graph seems to change in every section over the time and by that showing an instability.

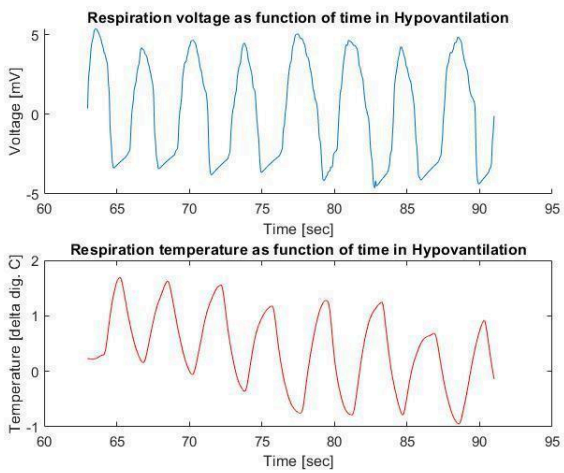

Figure 9: Voltage [mV] and Temperature [ΔC°] signals vs *time of Respiration test in hypoventilation*

As figure 9 above shows, an inverse relation of the respiratory voltage and the temperature's trends is apparent. Also, the voltage graph shows a quite moderate and constant breathing rate. On the contrary, the temperature graph seems to decrease in both its peaks levels and its minimum levels with the passing time.

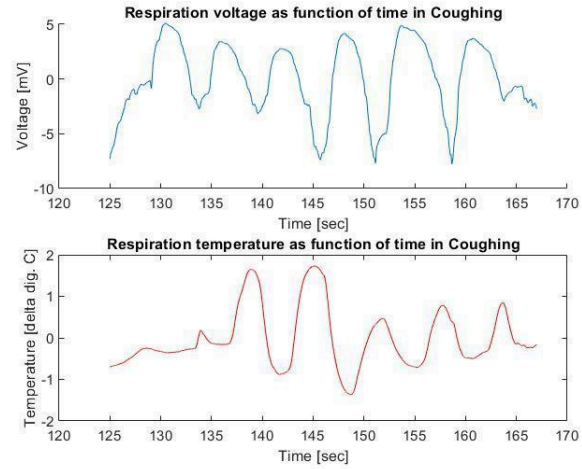

Figure 10: Voltage [mV] and Temperature [ΔC°] signals vs *time of Respiration test while coughing*

In figure 10 above the coughing activity cannot clearly be seen, but as the graph shows, in the beginning of the record a moderate breathing activity detected, and at around 143 seconds from the beginning of the test the breathing started to behave differently as the minimum voltage decreased dramatically ever since this point and until the end of the coughing segment record. On the other hand, the temperature graph doesn't seem to fit the voltage behavior as only 5 peaks appear for the temperature while 6 peaks appear for the voltage and the magnitude of the peaks along the record decreased in time while in the voltage graph the opposite happened.

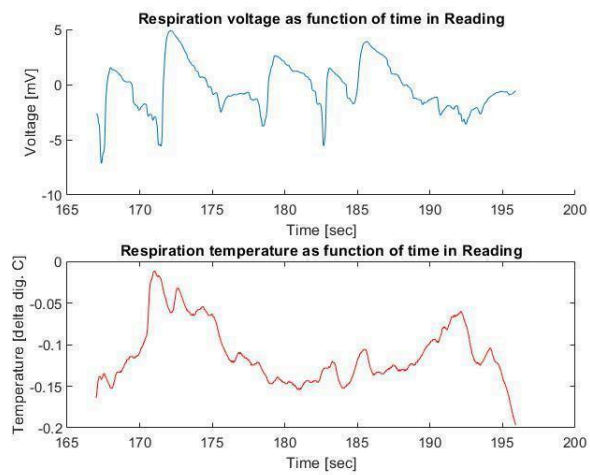

Figure 11: Voltage [mV] and Temperature [ΔC°] signals vs *time of Respiration test while reading*

Figure 11 shows a breathing while reading segment. As it shows, the respiratory voltage and the temperature are not stable and cyclic as in the previous graphs. Although, a quite stair-shaped graph is shown for the voltage graph.

Table 5: Comparison between the durations and the overall respiration rate of each segment in the respiration test

segment	Mean segment expiration Durations	STD segment expiration Durations	Mean segment inspiration Durations	STD segment inspiration Durations	Total segment Durations	RR value
Eupnea	2.76	0.07	2.24	0.04	15.00	15.99
Hyperven-tilation	1.00	0.035	0.88	0.13	30.17	33.81
Hypoventi-lation	1.40	0.25	2.17	0.34	24.94	19.25
Coughing	4.17	1.08	2.27	0.21	30.21	11.92
Reading	3.55	1.67	1.68	1.13	22.39	13.40

Table 5 above shows the average (formula 10) inspiration and expiration durations that calculated for the voltage of each segment of breathing functionalities (Eupnea, Hyperventilation, Hypoventilation, Coughing and Reading) , the standard deviation (formula 11) of these durations and also the overall respiration rate that was calculated for each segment. As it appears from table 5, the coughing segment was the one required for the most expiration and inspiration durations with the highest std parameter for the inspiration durations. While the hyperventilation had the lowest expiration and inspiration durations, with std of 0.3515 and 0.1362 respectively. In addition, the longest RR was the hyperventilation's segment and the lowest std was the eupnea's segments in both inspiration and expiration. The hyperventilation had the highest RR value with the lowest inspiration and expiration durations, so it had the highest frequency.

On the other hand, the lowest RR detected in the coughing segment, the highest std expiration detected in the coughing segment and the lowest RR was the coughing's segment. Also, the hypoventilation's RR value had also a high RR value with low expiration duration and high inspiration duration, so it had a moderate frequency.

3.1.3 Wavelet Coherence

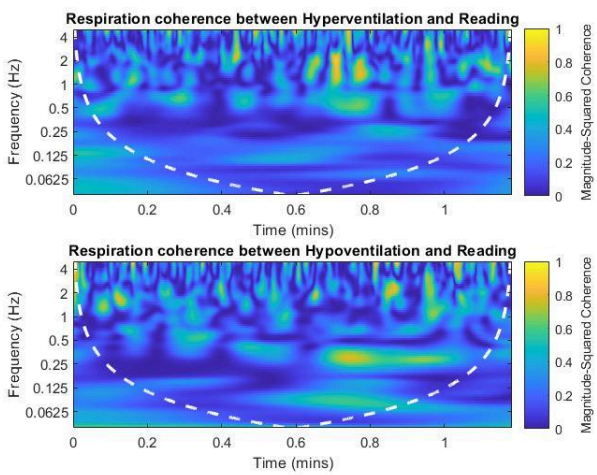

Figure 12: Wavelet Coherence graph of hyperventilation and hypoventilation vs reading-coughing signals in respiration test

From the comparison between the coherence of respiration's graph in hyperventilation and hypoventilation to the coughing-reading segment in figure 12 above several things can be understood. Firstly, the graph of both plots is limited to 6 Hz frequency since above this frequency only noises can be identified. Secondly, as can be seen in the hyperventilation's graph, a higher coherence magnitude is recognized for a high frequency (around 2 Hz). Meanwhile, a high coherence magnitude for low frequency (around 0.25 Hz) can be recognized at the hypoventilation's graph. When comparing these results to the results of table 5 It is clear that the coughing and reading has a pretty close RR value around 12 breaths/min and they both had a pretty close inspiration and expiration directions of around 4 and 2 respectively, meaning this segment had a pretty low overall frequency. For that reason, in the hypoventilation's coherence graph a wider yellow area can be seen for low frequency, while in the hyperventilation's graph a small yellow area appears for a higher frequency.

RR and PFT graphs

The Respiration Rate (RR) calculated, as described in the "Data Analysis and Values Calculations" section (formula 12), in order to expose the changing in the breathing rate over time. Graph 13 below, and the discussion afterwards compare the RR with the pulmonary functionality test (PFT) by the respiration voltage and temperature over time with reference to the difference between each segment and the expectations related.

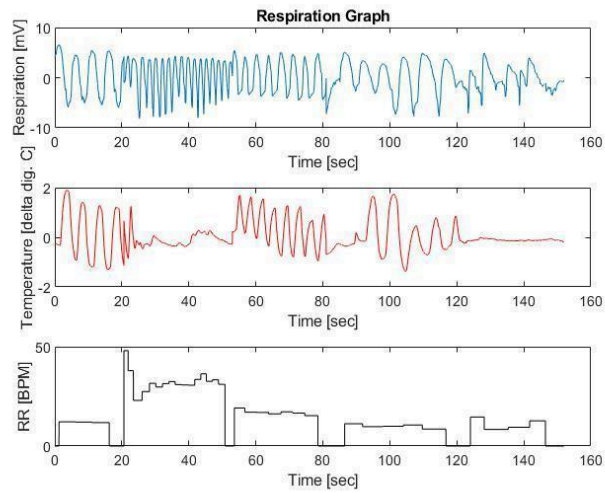

Figure 13: Respiration graphs of Voltage [mV], Temperature [ΔC°] and RR as function of time [sec]

Figure 13 above showing the RR graph that was calculated based on the respiration's voltage during the whole respiratory test. First, It can clearly seem that the segments are separated from each other. In the first segment - eupnea was performed and an almost flat RR value was detected. In the second segment hyperventilation was performed and high and not uniform RR values can be seen. In the next three other segments a moderate RR with close RR values detected in each segment - meaning the breathing rate was regular and moderate.

3.2 Part 3

Control group (spirometry test)

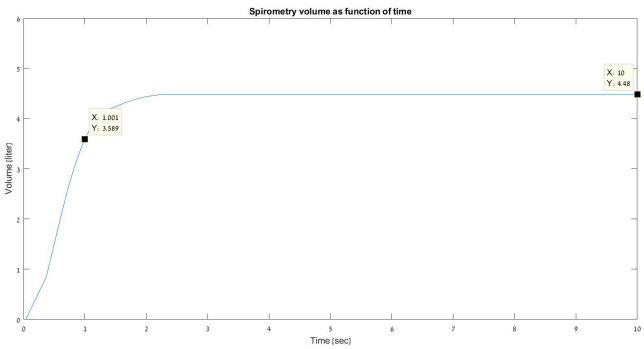

Figure 14: Spirometry volume as function of time (Control group)

It can seem at figure 14, that the volume increases until it gets to a plateau of 4.48 [L]. Moreover, after one second, the volume is 3.59 [L].

Comparison between the test and calculation

Tables 6: FEV1, FVC and Ratio FEV1/FVC values of control group, test group and theoretical values

	FEV1[L]	FVC[L]	FEV1%
Control Group	3.59	4.48	80.13
Test Group	3.42	4.41	77.55
Theoretical value	3.36	4.1	81.59

All the FEV1/FVC ratios are above 70%, when the control group and the test group are also lower than 80%.

Table 7: FEV1, FVC and Ratio FEV1/FVC errors

	E_{FEV1}[%]	E_{FVC}[%]	$E_{FEV1\%}$[%]
Control vs Test	4.76	1.56	3.22
Control vs Theoretical	6.85	9.27	1.79
Test vs Theoretical	1.79	7.56	4.95

Table 7 represents the errors between the values and the results of the groups which examined: control group, test group and theoretical values. It can seem that all the errors of FEV1/FVC ratios are less than 5%, meaning the diagnosis of COPD is the same. However, the errors of FEV1 and FVC values between control group and theoretical values are more than 5%. The error of FVC value between test group and theoretical value is also above 5%.

4. Discussion & Conclusion

4.1. Part 1

The respiratory result that is shown in the graph (figure 4) is compatible with the expectations, since air flow represents the dynamic movement of the air while breathing, and thus at the moment that the examinee starts exhaling the airflow graph immediately decreases while the air volume is still above the average breathing volume. On the other hand, the relative errors that were calculated for this part showed incoherence to the theoretical values as the RV relative error was 14.952% and the VC relative error was 47.236% and they both much higher than 5% which determine coherence. This incoherency can be explained by several different reasons. The RV couldn't be determined by the spirometer test and therefore a default value of 1 Liter was used in the Biopack. This default value is a general value and is not related to the examinee's breathing abilities. Thus, the theoretical RV value, which is calculated based on the examinee's gender, age and height, is not coherent to the default value. The theoretical VC value was much higher (around 1.6 Liter difference) then the value that was calculated from the spirometry test. That means that from the examinee's gender, height and age alone, she should have been capable of getting to greater breathing volumes than she actually performed in the experiment. But, breathing ability isn't determined only by the parameters of gender, height and age. The examinee might have breathing difficulty which did not come into account. In addition, it is probable that the examinee didn't make the most effort to inhale and exhale to her maximal ability and thus the relative error showed incoherence. Additionally, it can be clearly seen in the graph at figure 5 that the extreme exhalation did not perform well and the parameters from table 3 also reinforce that with a very low ERV volume and very close RV and FRC values and correspondingly also VC and IC has very close values. The disappearance of the extreme exhalation part probably affected the VC that was calculated in the experiment by lowering its actual value and thus provided a significant relative error compared to the theoretical VC volume.

The spectrogram graph came along with the expectation as the highest power detected for low frequencies under 2 Hz, and at the time the extreme inhale performed a high power detected over all the frequencies.

In addition, this part of the experiment includes examination in the frequency and time domains (figure 6). It can seem that the low frequencies are more dominant in breathing while resting. It is corresponding to the resting respiratory frequency range which is between 0.16 to 0.33 Hz [24]. In extreme inhale and exhale the high frequencies were more dominant. On the other hand, the time domain shows the trend of the breathing cycle and thus can better express the changes in the breathing behavior of the examinee over time. Accordingly, it can be concluded that in order to achieve information about the breathing frequency range the frequency-domain is the better choice for that.

4.2 Part 2

4.2.1. The suitability of the voltage and temperature results to the expectations

The respiration test, as mentioned, is divided into 5 segments of different breathing functionality. Since the inhalation is accompanied by insertion of cold air while exhalation is accompanied by ejection of hot air, when voltage shows increasing, which is caused by inhale, the temperature expected to show decrease, which is caused by cold air that is detected in the temperature sensor that located under the examinee's nostril. Thus, the temperature was expected to have an inverse trend to the voltage. As has been shown, almost all the segments' graphs fitted this expectation. In addition, most of the segment's graphs behaved as expected; in the eupnea segment, moderate trend with closely constant rate detected and assured with low std in both the expiration and inspiration durations. The hyperventilation was found with the highest RR and the lowest inspiration and expiration durations. Also, the reading segment had a reasonable stair-shaped graph with the highest std in both inspiration and expiration as the reading has an unsystematic trend. The non-systematic voltage and temperature in the reading segment makes sense as reading requires breathing in a pace that matches the reading pace, and

thus, the voltage nor the temperature is systematic. Besides, the reading temperature graph wasn't consistent with the reading behavior as while reading the breathing functionality disrupted and thus there is no systematic inhale and exhale through the nose so the temperature sensor, which is located under the nostril, doesn't detect synchronized breathing function. However, the hypoventilation's RR didn't come along with the expectation due to the fact that hypoventilation's RR value was not the lowest RR value in table 5 and bypassed all the RR values of all the other segments, except for the hyperventilation's RR. This contrast can be explained by the way that the experiment was performed. The examinee was not in a real hypoventilation, she tried to mimic the hypoventilation demeanor. Moreover, the hypoventilation segment came right after the hyperventilation segment and thus it makes sense to assume that the examinee tried to slow down her breathing rate relative to the hyperventilation rate. Furthermore, the coughing graph did not behave as expected due to not showing a clear sudden drop in the graph. The coughing graph shows a moderate resting breathing of 6 strokes and then a dramatic drop at the end of the 6th stroke that might be a result of coughing activity. But afterwards it seems like the coughing activity continued for the rest of the segment record as the maximum voltage doesn't change and the minimum voltage stays low as in the coughing drop. This phenomenon can be explained in many ways. First, maybe the coughing activity did proceed. Second, the coughing might have changed the way the examinee breathed as her breathing was different than previously. The explanation to the temperature behavior in the coughing segment, might be the fact that while coughing, most of the air comes out of the mouth and not from the nose, and thus the temperature didn't accompany the breathing as happened in the eupnea's segment for example. Also, it must be mentioned that the temperature measurement isn't accurate, and it is impossible to expect consistency in its behavior since the temperature sensor was held by the examinee during the experiment and it's not unfounded to assume that the examinee's hand moved during the recording.

4.2.2. Wavelet Coherence

Comparison between the coherence of respiration's graph in hyperventilation and hypoventilation to the coughing-reading segment in figure 12 showed adequacy to expectations. The hyperventilation's graph had a higher coherence magnitude for a high frequency around 2 Hz, and the hypoventilation had a high coherence magnitude for low frequency around 0.25 Hz. When comparing these results to the results of table 5, the results from the two measures fit. In the hypoventilation's coherence graph a wider yellow area can be seen for low frequency, as the hypoventilation had an inspiration duration which was very close to the coughing-reading inspiration duration. Also, the hypoventilation had a closer RR and expiration duration to the coughing-reading segment compared to the hyperventilation. Thus, the duration of high coherence with the hypoventilation was wider. While, in the hyperventilation's graph a small yellow area appears for a higher frequency, as it has a smaller suitability to the coughing-reading segment trends.

In addition, the subject's breathing pattern is examined in hyperventilation and hypoventilation. Based on the experiment and as described in the introduction, the respiratory centers are strongly affected by chemoreceptors. The chemoreceptors are activated by the pH levels in the blood. Low pH levels in the blood are resulted by high concentrations of carbon dioxide which occurs in hypoventilation. The activation of the chemoreceptors leads the respiratory centers in the brain to increase the rate and the depth of the ventilation. The low concentrations of carbon dioxide in the blood during hyperventilation cause much less stimulation of the chemoreceptors, and therefore, leads to decrease in ventilation and the respiratory ventilation will be slower and sufficient. After hypoventilation, the respiratory centers in the brain will try strongly to increase the ventilation, especially to eject carbon dioxide by exhalation, and thus, after hyperventilation it will be easier for a person to hold his breath.

4.2.3. RR and PFT graphs

The apparent separation of each segment in the RR graph happened since the RR value was calculated between two adjacent peaks, and as it could clearly see in the plots of figure 13 compared to the plots of the segments in the figures 7-11, the beginning and end of each segment had low voltage values, and thus those areas are not considered in the RR calculation. As the RR graph at figure 13 shows, most of the segment's respiration rates were moderate and quite stable. The eupnea RR segment looks almost constant as expected for resting breathing without any disruptions. In addition, the hypoventilation and the coughing segments unsurprisingly looked very moderate with almost constant RR value, by that it can assume that the coughing action performed in coherence to the breathing rate. The reading segment also showed a quite moderate RR trend and thus can be assumed that the reading performed with an almost constant rhythm. On the other hand, the hyperventilation, as also could clearly seen in figure 8 didn't keep on a constant voltage level and was changed all along the recording. This can be explained by the fact that hyperventilation is a phenomenon in which a person at an unstable phase and thus much probably couldn't keep on a moderate and constant breathing rate. Also, it shouldn't be forgotten that this experiment performed by an examinee who didn't was under a real hyperventilation situation and was only trying to mimic the hyperventilation kind of functionality. However, the hyperventilation had the highest RR values in the graph and also its stairs were thinner than those of all the other segments as also discussed about table 5 where the hyperventilation had the lowest inspiration and expiration durations as expected.

4.3 part 3

The trend shown in figure (14) is consistent with expectation - the maximum value of humans is close to 6 [L] when on average, RV is equal to 1 [L]. Thus, it makes sense that FVC will be close to 5 [L] [23]. Moreover, the plateau is another characteristic which portrays the results of this Spirometry test.

In addition, as expected, based on the data of both the spirometry test and mobile phone, it appears that the examinee belongs to normal values, meaning not suffering from COPD. The low error of all FEV1/FVC ratios indicate that the mobile phone manages to well diagnose unpresence of disease.

On the other hand, both FEV1 and FVC of theoretical values are lower than values of the control group and the test group. The more significant errors were seen for the control group against the theoretical value, but the total error of the ratio is the lowest because there is a suitability between the two values. Moreover, because the theoretical FVE1 value is closer to the test FVE1 value than the gap of FVC value between those groups, there is a significant difference between theoretical and

test group FVC value, and that causes an increase in the error of the ratio.

Improvement Recommendations

Although the mobile phone was able to correctly diagnose specific examinees (healthy female, 26 years old), the more important use-case is to check if the application can correctly diagnose a wide range of examinees at different ages, heights, races, and COPD levels. For future experiments, the Improvement recommendations is to apply the application on a diverse group of examinees. Furthermore, in order to save resources, the option should be examined for utilizing the results of the first experiment to extract the tested values in the third part of (FVE1 and FVC, control group)' for example calculating FVE1 after about a one second from the strenuous exhalation [18].

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